A Formal Proof of Partition Function in Lean 4

Mainak Bagui
Pondicherry University
mainakbagui2304@gmail.com

31st October 2026

Abstract

We present a formalization, in the Lean 4 proof assistant with the Mathlib library, of the partition function $p(n)$ and of four classical theorems about it: Euler's generating function, Ramanujan's congruence $p(5n+4) \equiv 0 \pmod 5$, the Hardy–Ramanujan asymptotic formula and Rademacher's exact series. Euler's identity $\sum_{n\geq 0} p(n)X^n = 1/(X;X)_\infty$ is proved completely, over an arbitrary field. Ramanujan's congruence is proved by his elementary argument relative to one classical statement that is formalized but not yet proved, namely Jacobi's identity for $(X;X)_\infty^3$. The two analytic theorems are stated formally, with Dedekind sums, the sums $A_k(n)$ and Rademacher's kernel defined in full; we prove the parts of their proofs that do not need the circle method (reality and the trivial bound of $A_k(n)$, the Bessel form of the terms, the saddle-point computation) and reduce Rademacher's theorem to two analytic statements. The development comprises about 950 lines and compiles against current Mathlib. Exactly ten statements remain unproved; we list them, together with numerical evidence for their correctness. The source is available at https://github.com/mainakbagui/A-Formal-Proof-of-Partition-Function-in-Lean-4.

Keywords: partition function, Lean 4, Mathlib, formalized mathematics, Ramanujan congruences, Rademacher series.
MSC 2020: 11P82, 11P83, 11F20, 05A17, 68V20.

1 Introduction

A *partition* of an integer $n \geq 0$ is a finite multiset of positive integers with sum n, and $p(n)$ denotes the number of partitions of n; thus $p(0) = 1$ and $p(4) = 5$. Four classical theorems dominate the theory of $p(n)$ [1, 2, 7]. Writing $(q;q)_\infty = \prod_{m\geq 1}(1-q^m)$, they are Euler's generating function

$$\sum_{n\geq 0} p(n)\, q^n = \frac{1}{(q;q)_\infty}, \tag{1.1}$$

Ramanujan's congruence [17]

$$p(5n+4) \equiv 0 \pmod 5 \qquad (n \geq 0), \tag{1.2}$$

the asymptotic formula of Hardy and Ramanujan [6]

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\!\left(\pi\sqrt{\tfrac{2n}{3}}\right) \qquad (n \to \infty), \tag{1.3}$$

and Rademacher's exact formula [15], valid for every $n \geq 1$,

$$p(n) = \frac{1}{\pi\sqrt{2}} \sum_{k=1}^{\infty} A_k(n)\, \sqrt{k}\, \frac{d}{dn}\left(\frac{\sinh\!\left(\frac{\pi}{k}\sqrt{\frac{2}{3}\left(n-\frac{1}{24}\right)}\,\right)}{\sqrt{n-\frac{1}{24}}} \right), \tag{1.4}$$

where $A_k(n) = \sum_{0 \le h < k,\ \gcd(h,k)=1} e^{\pi i s(h,k) - 2\pi i n h/k}$ and $s(h,k)$ is a Dedekind sum (Section 6). This paper reports on a formalization of these four results in Lean 4 [14] on top of Mathlib [13]. All definitions and all statements are formalized. The proofs are formalized to the following extent.

- *Generating function.* The identity (1.1) is proved completely, as an identity of formal power series over an arbitrary field, and in the inverse-free form $\left(\sum_n p(n)X^n\right)(X;X)_\infty = 1$ over an arbitrary commutative ring (Theorem 3.3).
- *Ramanujan's congruence.* The congruence (1.2) is proved by Ramanujan's elementary argument, relative to a single statement that we formalize but do not prove: Jacobi's identity $(X;X)_\infty^3 = \sum_{k\ge 0}(-1)^k(2k+1)X^{k(k+1)/2}$ (Section 4).
- *Hardy–Ramanujan formula.* The formula (1.3) is stated. We also state the two analytic inputs of its proof by the circle method, and we prove the saddle-point computation that fixes the radius of the circle (Section 5).
- *Rademacher's series.* Dedekind sums, the sums $A_k(n)$ and the terms of (1.4) are defined in full. We prove that $A_k(n)$ is real, that $|A_k(n)| \le k$, that the differentiated kernel has the closed form used in the definition, and that the terms coincide with their expression through the Bessel function $I_{3/2}$. The theorem itself is proved relative to two analytic statements: a bound $O(k^{-3/2})$ for the k-th term, and the evaluation of the sum (Section 6).

Status of the formalization. Throughout, *proved* means accepted by the Lean kernel without any use of the placeholder `sorry`; *stated* means that the formal statement type-checks while its proof is `sorry`; and *proved relative to* a statement S means that the formal proof is complete except that it invokes the stated but unproved S. The development is a single file, `PartitionFunction.lean`, of about 950 lines, half of which are documentation. It contains 14 definitions and 50 theorems: 33 theorems are proved, 7 are proved relative to stated ones, and 10 are stated. The latter are listed in Table 2. Consequently only (1.1) is proved unconditionally at present, (1.2) depends on one classical identity, and (1.3) and (1.4) are formal statements accompanied by verified parts of their proofs. We do not claim complete formal proofs of (1.2)–(1.4). The purpose of this paper is to document what has been machine-checked, the design of the formal statements, and what remains to be done.

Related work. In Isabelle/HOL, Eberl formalized $p(n)$, its generating function and the pentagonal number theorem [3]; Eberl, Bordg, Paulson and Li formalized Dedekind sums [4]; and Eberl and Paulson formalized Rademacher's series with bounds for its remainder, and deduced (1.3) from it [5]. The latter shows that complete formal proofs of (1.3) and (1.4) are feasible, and it is the natural guide for completing the present development. In Lean, Mathlib contains partitions and their generating functions, the pentagonal number theorem, and the Dedekind eta function with its transformation under $\tau \mapsto -1/\tau$; our development builds on these. Lau, Lee and Ono [8] formalized the Jacobi triple product and the Rogers–Ramanujan identities in Lean; the triple product is the classical source of Jacobi's identity, which is our missing ingredient for (1.2). Another recent formalization of a classical theorem of number theory in Lean 4 is the irrationality of $\zeta(3)$ by Liu, Zhang and Zhi [12].

Organisation. Section 2 recalls the conventions of Lean and Mathlib that are needed to read the formal statements. Sections 3–6 treat the four theorems in turn, each with its informal statement, its formal statement and the formalized part of its proof. Section 7 gives the exact verification status and Section 8 the remaining work.

2 Lean Preliminaries

Lean 4 [14] is a proof assistant based on dependent type theory, and Mathlib [13] is its community-maintained library of mathematics. We used Lean `v4.36.0-rc2` and Mathlib at commit `3881f16c6c9b` (10 November 2026). The following conventions are needed to read the formal statements.

- *Types.* The type $\mathbb{N}$ contains 0, and `(n : ℝ)` is the image of `n : ℕ` in $\mathbb{R}$. The ring of formal power series over `R` is `R⟦X⟧`; the series $\sum_n f(n)X^n$ is `PowerSeries.mk f`, and `PowerSeries.expand p hp f` is $f(X^p)$, where `hp` is a proof of $p \neq 0$.
- *Total functions.* All functions are total: `x / 0 = 0`, the real square root `√x` is 0 for $x < 0$, and `f⁻¹ = 0` for a power series f over a field with constant coefficient 0. Hypotheses such as $n > 0$ in Theorem 6.3 are therefore part of the mathematical content of a statement.
- *Sums and products.* `HasSum f a` and `HasProd f a` assert that the sums, respectively products, of f over finite sets converge to a along the directed set of all finite subsets of the index type. For real and complex series this is absolute convergence. `Σ' i, f i` and `Π' i, f i` denote the limit if it exists, and 0, respectively 1, otherwise.
- *Asymptotics.* `u ~[l] v` means $u - v = o(v)$ along the filter `l`; `atTop` is the filter $n \to \infty$ and $\mathcal{N}$`[>] 0` the filter $t \to 0^+$.
- *Unproved statements.* A proof may be replaced by `sorry`. Lean then accepts the declaration with a warning and records its dependence on the axiom `sorryAx`. The commands `#print axioms` and `#print sorries` report, for a given theorem, the axioms and the unproved statements on which its proof depends. In our development every unproved statement is a separate named theorem, and no definition contains `sorry`.

From Mathlib we use the theory of partitions and of their generating functions, the pentagonal number theorem, the product topology on power series, the Frobenius endomorphism of power series, and the Dedekind eta function. The headers of the corresponding Mathlib files credit B. Mehta, W. Wang, A. Chambert-Loir, M. I. de Frutos-Fernández, W. Zou, C. Birkbeck and D. Loeffler.

3 The Generating Function

Mathlib defines `Nat.Partition n` as the type of multisets of positive integers with sum n. This type is finite, and $p(n)$ is its cardinality:

```
def partitionFunction (n : ℕ) : ℕ := Fintype.card (Nat.Partition n)
```

For a commutative ring R we consider the power series $E_R = (X;X)_\infty = \prod_{i\geq 1}(1 - X^i)$ and $P_R = \sum_{n\geq 0} p(n)X^n$ in $R[[X]]$. Their formal definitions are:

```
def qPochhammerInfinite (R : Type*) [CommRing R] : R⟦X⟧ :=
  letI : TopologicalSpace R := ⊥
  Π' i : ℕ, ((1 : R⟦X⟧) - X ^ (i + 1))

def partitionGenFun (R : Type*) [CommSemiring R] : R⟦X⟧ :=
  PowerSeries.mk fun n => (partitionFunction n : R)
```

Remark 3.1. An infinite product needs a topology. Mathlib equips `R⟦X⟧` with the topology of coefficientwise convergence induced by a topology on R. Inside the definition we choose the discrete topology on R (written $\bot$), which gives the X-adic topology. This choice is not visible from outside: we prove (`hasProd_qPochhammerInfinite`) that for every topology on R the partial products of $\prod_{i\geq 1}(1 - X^i)$ converge to `qPochhammerInfinite R`.

Theorem 3.2 (Euler's pentagonal number theorem)**.** *For every commutative ring R,*

$$E_R = \sum_{k\in\mathbb{Z}} (-1)^k X^{k(3k-1)/2}.$$

This theorem is in Mathlib (`PowerSeries.WithPiTopology.tprod_one_sub_X_pow`); in our notation it is `qPochhammerInfinite_eq_pentagonalSeries`.

Theorem 3.3 (Euler)**.** *For every commutative ring R we have $P_R \cdot E_R = 1$ in $R[[X]]$. Consequently $P_K = E_K^{-1}$ for every field K.*

Proof. For an arbitrary predicate on the positive integers, Mathlib proves that the generating function of the partitions all of whose parts satisfy the predicate is the product of the geometric series $\sum_{j\geq 0} X^{ij}$ over the admissible parts i (`Nat.Partition.hasProd_powerSeriesMk_card_restricted`); this is the formal counterpart of Euler's bijection between partitions and multiplicity vectors. For the trivial predicate it gives $P_R = \prod_{i\geq 1}\sum_{j\geq 0} X^{ij}$. We work in the X-adic topology, in which $R[[X]]$ is a Hausdorff topological ring. The termwise product of two convergent infinite products converges to the product of their values, so $\prod_{i\geq 1}(\sum_{j\geq 0} X^{ij})(1-X^i)$ converges to $P_R \cdot E_R$. Each factor of this product equals 1, because $(\sum_{j\geq 0} f^j)(1-f) = 1$ for every power series f with constant coefficient 0. Hence the constant family 1 has the two products $P_R \cdot E_R$ and 1, and these agree by uniqueness of limits. The second assertion follows because E_K has constant coefficient 1. ∎

The formal proof follows this argument line by line:

```
theorem partitionGenFun_mul_qPochhammerInfinite (R : Type*) [CommRing R] :
    partitionGenFun R * qPochhammerInfinite R = 1 := by
  let _ : TopologicalSpace R := ⊥
  have _ : DiscreteTopology R := ⟨rfl⟩
  have h := (hasProd_partitionGenFun R).mul (hasProd_qPochhammerInfinite R)
  have h1 : (fun i : ℕ => (Σ' j : ℕ, (X : R⟦X⟧) ^ ((i + 1) * j)) *
      (1 - X ^ (i + 1))) = fun _ => 1 := by
    funext i
    simp_rw [pow_mul]
    exact PowerSeries.WithPiTopology.tsum_pow_mul_one_sub_of_constantCoeff_eq_zero
      (by simp)
  rw [h1] at h
  exact h.unique hasProd_one
theorem partition_generating_function (K : Type*) [Field K] :
    partitionGenFun K = (qPochhammerInfinite K)⁻¹ := by
  rw [PowerSeries.eq_inv_iff_mul_eq_one]
  · exact partitionGenFun_mul_qPochhammerInfinite K
  · simp [constantCoeff_qPochhammerInfinite]
```

Combined with Theorem 3.2, Theorem 3.3 gives Euler's recurrence for $p(n)$ in the form $P_R \cdot \sum_{k\in\mathbb{Z}}(-1)^k X^{k(3k-1)/2} = 1$ (`partitionGenFun_mul_pentagonalSeries`).

Analytic form. The analytic theorems need (1.1) as an identity between holomorphic functions on the unit disc. Mathlib defines Euler's function ϕ (`eulerFunction`) as the sum of the pentagonal series and proves $\phi(q) = \prod_{n\geq 1}(1-q^n)$ for $|q| < 1$. The statement $\sum_{n\geq 0} p(n)q^n = 1/\phi(q)$ is the first one that we leave unproved:

```
theorem hasSum_partitionFunction_mul_pow {q : ℂ} (hq : ‖q‖ < 1) :
    HasSum (fun n : ℕ => (partitionFunction n : ℂ) * q ^ n)
      (eulerFunction q)⁻¹ := by
  sorry
```

It follows from Theorem 3.3 by evaluation at q, once absolute convergence is established. Relative to it we prove that $\sum_{n\geq 0} p(n)q^n = q^{1/24}/\eta(\tau)$ for $q = e^{2\pi i\tau}$ and $\operatorname{Im}\tau > 0$, where η is the Dedekind eta function of Mathlib and $q^{1/24} = e^{2\pi i\tau/24}$ (`hasSum_partitionFunction_qParam`).

4 Ramanujan's Congruence Modulo 5

Theorem 4.1 (Ramanujan [17]). *For every integer $n \geq 0$, $p(5n+4) \equiv 0 \pmod 5$.*

The formal statement in Lean 4 is:

```
theorem ramanujan_congruence_5 (n : ℕ) : partitionFunction (5 * n + 4) ≡ 0 [MOD 5]
```

We follow Ramanujan's argument as presented in [7, Ch. XIX]. Let $E = E_{\mathbb{F}_5}$ and $P = P_{\mathbb{F}_5}$, so that $PE = 1$ by Theorem 3.3.

Lemma 4.2 (Frobenius). *For every $f \in \mathbb{F}_5[[X]]$ we have $f(X)^5 = f(X^5)$. Consequently $P = E^4 \cdot P(X^5)$.*

Proof. The first assertion is a consequence of Mathlib's `PowerSeries.map_frobenius_expand`. Substituting $X \mapsto X^5$ in $PE = 1$ and using the first assertion gives $P(X^5)\,E^5 = 1$, hence $P = P \cdot P(X^5)E^5 = E^4P(X^5) \cdot PE = E^4P(X^5)$. ∎

```
theorem pow_five_eq_expand (f : (ZMod 5)⟦X⟧) :
    f ^ 5 = PowerSeries.expand 5 (by norm_num) f
theorem partitionGenFun_zmod_five_eq :
    partitionGenFun (ZMod 5) = qPochhammerInfinite (ZMod 5) ^ 4 *
      PowerSeries.expand 5 (by norm_num) (partitionGenFun (ZMod 5))
```

Lemma 4.3 (Jacobi's identity). *For every commutative ring R,*

$$E_R^3 = \sum_{k\geq 0} (-1)^k (2k+1)\, X^{k(k+1)/2}.$$

This is the only ingredient of Theorem 4.1 that is stated but not proved. The coefficient of X^n on the right-hand side is written as a finite sum over $k \leq n$ with at most one non-zero term:

```
def jacobiCubeSeries (R : Type*) [CommRing R] : R⟦X⟧ :=
  PowerSeries.mk fun n => Σ k ∈ Finset.range (n + 1),
    if k * (k + 1) / 2 = n then (-1) ^ k * (2 * (k : R) + 1) else 0
theorem qPochhammerInfinite_pow_three (R : Type*) [CommRing R] :
    qPochhammerInfinite R ^ 3 = jacobiCubeSeries R := by
  sorry
```

Classically, Lemma 4.3 is derived from the Jacobi triple product [1, 7].

Lemma 4.4. *Let $m \in \mathbb{Z}$ and $k, n \in \mathbb{N}$ satisfy $\frac{m(3m-1)}{2} + \frac{k(k+1)}{2} = 5n+4$. Then $2k+1 \equiv 0 \pmod 5$.*

Proof. The hypothesis is equivalent to $(6m-1)^2 + 3(2k+1)^2 = 24(5n+4)+4 = 5(24n+20)$. The squares modulo 5 are 0 and ± 1, so $x^2 + 3y^2 \equiv 0 \pmod 5$ forces $y \equiv 0 \pmod 5$; in other words, -3 is a quadratic non-residue modulo 5. In Lean the identity is verified by `linear_combination`, and the last step is decided by enumeration in `ZMod 5`. ∎

```
theorem two_mul_add_one_eq_zero_of_pentagonal_add_triangular (m : ℤ) (k n : ℕ)
    (h : pentagonal m + k * (k + 1) / 2 = 5 * n + 4) : 2 * (k : ZMod 5) + 1 = 0
theorem coeff_qPochhammerInfinite_pow_four (n : ℕ) :
    coeff (5 * n + 4) (qPochhammerInfinite (ZMod 5) ^ 4) = 0
```

Proof of Theorem 4.1 relative to Lemma 4.3. By Theorem 3.2 and Lemma 4.3,

$$E^4 = E \cdot E^3 = \sum_{m\in\mathbb{Z}} \sum_{k\geq 0} (-1)^{m+k}(2k+1)\, X^{m(3m-1)/2+k(k+1)/2}.$$

By Lemma 4.4, every term whose exponent is congruent to 4 modulo 5 has a coefficient divisible by 5. Hence the coefficient of X^{5n+4} in E^4 vanishes for every n (`coeff_qPochhammerInfinite_pow_four`). In the identity $P = E^4 \cdot P(X^5)$ of Lemma 4.2 the second factor involves only exponents divisible by 5. Therefore the coefficient of X^{5n+4} in P is a combination of the coefficients of $X^{5n'+4}$ in E^4 with $n' \leq n$, and it vanishes. ∎

Remark 4.5. Ramanujan's identity $\sum_{n\geq 0} p(5n+4)q^n = 5\,(q^5;q^5)_\infty^5/(q;q)_\infty^6$ exhibits the factor 5 explicitly. We state it without denominators in $\mathbb{Z}[[X]]$. It is not proved and it is not used.

```
theorem ramanujan_identity_five :
    (PowerSeries.mk fun n => (partitionFunction (5 * n + 4) : ℤ)) *
        qPochhammerInfinite ℤ ^ 6 =
      5 * PowerSeries.expand 5 (by norm_num) (qPochhammerInfinite ℤ) ^ 5 := by
  sorry
```

5 The Hardy–Ramanujan Asymptotic Formula

Theorem 5.1 (Hardy–Ramanujan [6])**.** *As $n \to \infty$, $p(n) \sim \dfrac{1}{4n\sqrt{3}} \exp\!\Big(\pi\sqrt{\dfrac{2n}{3}}\Big)$.*

The formal statement in Lean 4, which is not proved, is:

```
theorem hardy_ramanujan_asymptotic :
    (fun n : ℕ => (partitionFunction n : ℝ)) ~[atTop]
      (fun n : ℕ => 1 / (4 * (n : ℝ) * √3) *
        Real.exp (π * √(2 * (n : ℝ) / 3))) := by
  sorry
```

Relative to it, we prove the equivalent formulation that the quotient of the two sides tends to 1 (`hardy_ramanujan_tendsto`).

The circle method. Let $P(q) = \sum_{n\geq 0} p(n)q^n = 1/\phi(q)$ for $|q| < 1$. The proof of Theorem 5.1 has four steps [6, 1].

(i) By Cauchy's formula, $p(n) = \frac{1}{2\pi i} \oint_{|z|=r} \phi(z)^{-1} z^{-n-1}\, dz$ for $0 < r < 1$.

(ii) The transformation $\eta(-1/\tau) = \sqrt{-i\tau}\,\eta(\tau)$ gives, for $t > 0$,

$$P(e^{-t}) = \sqrt{\frac{t}{2\pi}}\, \exp\!\Big(\frac{\pi^2}{6t} - \frac{t}{24}\Big)\, P(e^{-4\pi^2/t}), \tag{5.1}$$

hence $P(e^{-t}) \sim \sqrt{t/2\pi}\, e^{\pi^2/6t}$ as $t \to 0^+$.

(iii) Near $z = e^{-t}$ the integrand on the circle $|z| = e^{-t}$ is thus $\exp(\pi^2/6t + nt)$, up to a factor of polynomial size, and t is chosen so as to minimize this quantity.

(iv) The arc near $z = 1$ yields the main term. The rest of the circle contributes $O\big(e^{c\pi\sqrt{2n/3}}\big)$ for some constant $c < 1$.

The transformation of η in step (ii) is available in Mathlib; we restate it pointwise as `eta_S_transform`, which is proved. Steps (i) and (ii) are stated but not proved:

```
theorem partitionFunction_eq_circleIntegral (n : ℕ) {r : ℝ}
    (hr0 : 0 < r) (hr1 : r < 1) :
    (partitionFunction n : ℂ) =
      (2 * π * Complex.I)⁻¹ *
        ∮ z in C(0, r), (eulerFunction z)⁻¹ / z ^ (n + 1) := by
  sorry
theorem partitionGenFun_asymptotic_near_one :
    (fun t : ℝ => Σ' n : ℕ, (partitionFunction n : ℝ) * Real.exp (-(n : ℝ) * t))
      ~[𝓝[>] 0]
      (fun t : ℝ => √(t / (2 * π)) * Real.exp (π ^ 2 / (6 * t))) := by
  sorry
```

Step (iii) is proved:

Lemma 5.2 (Saddle point)**.** *Let $n \in \mathbb{N}$ and $t > 0$. Then $\pi\sqrt{2n/3} \le \dfrac{\pi^2}{6t} + nt$, with equality for $t = t_0 = \pi/\sqrt{6n}$ if $n \ge 1$.*

Proof. By the inequality of arithmetic and geometric means, $\left(\frac{\pi^2}{6t} + nt\right)^2 \ge 4 \cdot \frac{\pi^2}{6t} \cdot nt = \frac{2\pi^2 n}{3}$. The value at t_0 is computed directly. ∎

```
theorem saddle_point_minimal (n : ℕ) {t : ℝ} (ht : 0 < t) :
    π * √(2 * (n : ℝ) / 3) ≤ π ^ 2 / (6 * t) + n * t
theorem saddle_point_value (n : ℕ) (hn : 0 < n) :
    π ^ 2 / (6 * (π / √(6 * (n : ℝ)))) + n * (π / √(6 * (n : ℝ))) =
      π * √(2 * (n : ℝ) / 3)
```

Thus the circle of integration has radius $e^{-\pi/\sqrt{6n}}$, and the exponent $\pi\sqrt{2n/3}$ of (1.3) is the minimal value in Lemma 5.2.

6 Rademacher's Exact Series

6.1 Dedekind sums and the sums $A_k(n)$

Let $((x)) = x - \lfloor x \rfloor - \frac{1}{2}$ for $x \in \mathbb{R} \setminus \mathbb{Z}$ and $((x)) = 0$ for $x \in \mathbb{Z}$. For $h \in \mathbb{Z}$ and an integer $k \ge 1$, the *Dedekind sum* is $s(h,k) = \sum_{r \bmod k} ((r/k))((hr/k))$. We put $\omega_{h,k} = e^{\pi i s(h,k)}$ and

$$A_k(n) = \sum_{\substack{0 \le h < k \\ \gcd(h,k)=1}} \omega_{h,k}\, e^{-2\pi i n h/k}.$$

```
def sawtooth (x : ℝ) : ℝ := if Int.fract x = 0 then 0 else Int.fract x - 1 / 2
def dedekindSum (h : ℤ) (k : ℕ) : ℝ :=
  Σ r ∈ Finset.range k, sawtooth ((r : ℝ) / k) * sawtooth ((h : ℝ) * r / k)
def rademacherOmega (h : ℤ) (k : ℕ) : ℂ :=
  Complex.exp (((π * dedekindSum h k : ℝ) : ℂ) * Complex.I)
def rademacherSumComplex (k n : ℕ) : ℂ :=
  Σ h ∈ (Finset.range k).filter (fun h => Nat.Coprime h k),
    rademacherOmega (h : ℤ) k *
      Complex.exp (((-(2 * π * n * h / k) : ℝ) : ℂ) * Complex.I)
def rademacherSum (k : ℕ) (n : ℕ) : ℝ := (rademacherSumComplex k n).re
```

The sum $A_k(n)$ is defined by complex exponentials, but it enters (1.4) as a real number. We define `rademacherSum` as the real part of the complex sum and prove that nothing is lost.

Proposition 6.1. *Let $h \in \mathbb{Z}$, $k \ge 1$ and $n \ge 0$.*

(a) *$s(-h,k) = -s(h,k)$ and $s(h+k,k) = s(h,k)$; hence $s(k-h,k) = -s(h,k)$.*

(b) $A_k(n)$ *is real, and* $A_k(n) = \sum_{0 \le h < k,\ \gcd(h,k)=1} \cos(\pi s(h,k) - 2\pi n h/k)$.

(c) $|A_k(n)| \le k$, *and* $A_1(n) = 1$.

Proof. (a) follows from $((-x)) = -((x))$ and $((x+m)) = ((x))$ for $m \in \mathbb{Z}$. For (b), let $\theta_h = \pi s(h,k) - 2\pi n h/k$, so that the imaginary part of $A_k(n)$ is $\sum_h \sin\theta_h$. If $k \ge 2$, then $h \mapsto k-h$ is an involution of the set of indices, and $\theta_{k-h} = -\theta_h - 2\pi n$ by (a). Hence $\sum_h \sin\theta_h$ equals its own negative. The case $k = 1$ is immediate, since the only index is $h = 0$ and $s(0,1) = 0$. (c) follows from (b). ∎

```
theorem rademacherSumComplex_im (k n : ℕ) : (rademacherSumComplex k n).im = 0
theorem rademacherSum_eq_sum_cos (k n : ℕ) :
    rademacherSum k n = Σ h ∈ (Finset.range k).filter (fun h => Nat.Coprime h k),
      Real.cos (π * dedekindSum (h : ℤ) k - 2 * π * n * h / k)
theorem abs_rademacherSum_le (k n : ℕ) : |rademacherSum k n| ≤ k
```

6.2 The terms of the series

For $k \ge 1$ and $x > \frac{1}{24}$ let $F_k(x) = \sinh\!\big(\frac{\pi}{k}\sqrt{\frac{2}{3}(x-\frac{1}{24})}\,\big)/\sqrt{x-\frac{1}{24}}$. The definition of the terms of (1.4) contains the derivative of F_k in closed form, so that it does not depend on the derivative operator: with $u = x - \frac{1}{24}$ and $y = \frac{\pi}{k}\sqrt{\frac{2}{3}u}$,

$$F_k'(x) = \frac{\pi\sqrt{2/3}}{k}\cdot\frac{\cosh y}{2u} - \frac{\sinh y}{2u\sqrt{u}} = \frac{y\cosh y - \sinh y}{2u\sqrt{u}}. \tag{6.1}$$

We prove that the closed form is the derivative, and define the k-th term of (1.4) by

$$T_k(n) = \frac{1}{\pi\sqrt{2}}\,A_k(n)\,\sqrt{k}\,F_k'(n). \tag{6.2}$$

```
def rademacherKernel (k : ℕ) (x : ℝ) : ℝ :=
  Real.sinh (π / k * √(2 / 3 * (x - 1 / 24))) / √(x - 1 / 24)
def rademacherKernelDeriv (k : ℕ) (x : ℝ) : ℝ :=
  π * √(2 / 3) / k * Real.cosh (π / k * √(2 / 3 * (x - 1 / 24))) /
      (2 * (x - 1 / 24)) -
    Real.sinh (π / k * √(2 / 3 * (x - 1 / 24))) /
      (2 * (x - 1 / 24) * √(x - 1 / 24))
theorem hasDerivAt_rademacherKernel (k : ℕ) {x : ℝ} (hx : 1 / 24 < x) :
    HasDerivAt (rademacherKernel k) (rademacherKernelDeriv k x) x
def rademacherTerm (k : ℕ) (n : ℕ) : ℝ :=
  1 / (π * √2) * rademacherSum k n * √(k : ℝ) * rademacherKernelDeriv k n
```

Proposition 6.2 (Bessel form)**.** *Let* $I_{3/2}(z) = \sqrt{2/(\pi z)}\,(\cosh z - \frac{\sinh z}{z})$ *for* $z > 0$. *For all integers* $k, n \ge 1$,

$$T_k(n) = \frac{2\pi}{(24n-1)^{3/4}}\cdot\frac{A_k(n)}{k}\cdot I_{3/2}\Big(\frac{\pi\sqrt{24n-1}}{6k}\Big).$$

Mathlib contains the Bessel function J_a of the first kind but not yet the modified function I_ν, so $I_{3/2}$ is defined by its closed form. Proposition 6.2 is proved; the formal proof expresses every radical through $\sqrt{2}$, $\sqrt{3}$, $\sqrt{k}$, $w = \sqrt{24n-1}$ and $\sqrt{w}$, and concludes with the tactics `field_simp` and `ring`.

```
def besselIThreeHalves (z : ℝ) : ℝ :=
  √(2 / (π * z)) * (Real.cosh z - Real.sinh z / z)
theorem rademacherTerm_eq_bessel (k n : ℕ) (hk : 0 < k) (hn : 0 < n) :
    rademacherTerm k n =
      2 * π / (24 * (n : ℝ) - 1) ^ (3 / 4 : ℝ) * (rademacherSum k n / k) *
        besselIThreeHalves (π * √(24 * (n : ℝ) - 1) / (6 * k))
```

6.3 The main theorem

Theorem 6.3 (Rademacher [15]). *For every integer $n \geq 1$ the series $\sum_{k\geq 1} T_k(n)$ converges absolutely, and its sum is $p(n)$.*

The formal statement uses `HasSum`, with the index shifted so that `k : ℕ` corresponds to the term $k+1$. Its proof is a reduction to two statements that are not proved:

```
theorem abs_rademacherTerm_le (n : ℕ) (hn : 0 < n) :
    ∃ C : ℝ, ∀ k : ℕ, 0 < k →
      |rademacherTerm k n| ≤ C / (k : ℝ) ^ (3 / 2 : ℝ) := by
  sorry
theorem tsum_rademacherTerm (n : ℕ) (hn : 0 < n) :
    Σ' k : ℕ, rademacherTerm (k + 1) n = (partitionFunction n : ℝ) := by
  sorry
theorem rademacher_exact_series (n : ℕ) (hn : n > 0) :
    HasSum (fun k : ℕ => rademacherTerm (k + 1) n) (partitionFunction n : ℝ) :=
  (summable_rademacherTerm n hn).hasSum_iff.mpr (tsum_rademacherTerm n hn)
```

Here `summable_rademacherTerm` is proved relative to the first statement, by comparison with $\sum_k k^{-3/2}$. The first statement is elementary. Indeed $0 \leq y\cosh y - \sinh y \leq y^3 \cosh y$ for $y \geq 0$, so that (6.1) gives $F_k'(n) = O_n(k^{-3})$, and $|A_k(n)|\sqrt{k} \leq k^{3/2}$ by Proposition 6.1. The second statement is the analytic core of Rademacher's theorem.

The analytic core. In the proof [15, 16, 2] one applies Cauchy's formula on a path that depends on a parameter N: the circle $|q| = e^{-2\pi/N^2}$ dissected by the Farey fractions of order N or, in the later version, the Ford circles in the τ-plane. Near the root of unity $e^{2\pi ih/k}$ one uses the transformation of η under an element of $\mathrm{SL}_2(\mathbb{Z})$ that maps h/k to $i\infty$. We state the general transformation law in the form of [2, Theorem 3.4]: for $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ with $c > 0$ and $\operatorname{Im}\tau > 0$,

$$\eta\Big(\frac{a\tau+b}{c\tau+d}\Big) = \exp\Big(\pi i\Big(\frac{a+d}{12c} + s(-d,c)\Big)\Big)\,\big(-i(c\tau+d)\big)^{1/2}\,\eta(\tau), \tag{6.3}$$

with the principal branch of the square root.

```
theorem eta_modular_transformation (a b d : ℤ) (c : ℕ) (hc : 0 < c)
    (hdet : a * d - b * c = 1) {τ : ℂ} (hτ : 0 < τ.im) :
    ModularForm.eta ((a * τ + b) / (c * τ + d)) =
      Complex.exp (π * Complex.I *
          (((a + d : ℤ) : ℂ) / (12 * c) + ((dedekindSum (-d) c : ℝ) : ℂ))) *
        Complex.sqrt (-Complex.I * (c * τ + d)) * ModularForm.eta τ := by
  sorry
```

Mathlib provides the case $(a,b,c,d) = (0,-1,1,0)$. By (6.3), the function P near $e^{2\pi ih/k}$ equals $\omega_{h,k}$ times an elementary factor times $1 + O(\text{exponentially small})$. The elementary factor integrates to the Bessel function of Proposition 6.2, the sum over h assembles $A_k(n)$, and the error terms are $O(N^{-1/2})$; letting $N \to \infty$ gives the second statement.

Remark 6.4. The series obtained by Hardy and Ramanujan in [6] is asymptotic but divergent, as shown by Lehmer [9]; Rademacher's modification of the kernel makes it convergent. The sums $A_k(n)$ and the remainder of Rademacher's series were studied in detail by Lehmer [10, 11]. The first term $T_1(n)$ is asymptotic to the right-hand side of (1.3); we state this as `rademacherTerm_one_isEquivalent` without proof. Together with a bound for the terms with $k \geq 2$ it yields the deduction of Theorem 5.1 from Theorem 6.3.

7 Verification Status

The file compiles without errors; the only warnings are the ten warnings for the use of `sorry`. Table 1 gives the status of the four main theorems and Table 2 lists the ten unproved statements

(O1)–(O10). All of them are classical theorems or routine consequences of classical theorems; none is a conjecture.

Result	Lean theorem	Status	Depends on
(1.1), Theorem 3.3	`partition_generating_function`	proved	—
(1.2), Theorem 4.1	`ramanujan_congruence_5`	proved relative to	(O2)
(1.3), Theorem 5.1	`hardy_ramanujan_asymptotic`	stated	is (O4)
(1.4), Theorem 6.3	`rademacher_exact_series`	proved relative to	(O8), (O9)

Table 1: Status of the four main theorems.

	Lean theorem	Content, and what a proof requires
(O1)	`hasSum_partitionFunction_mul_pow`	$\sum_n p(n)q^n = 1/\phi(q)$ for $\lvert q\rvert < 1$. Evaluation of Theorem 3.3; absolute convergence.
(O2)	`qPochhammerInfinite_pow_three`	Jacobi's identity (Lemma 4.3). Jacobi triple product, formalized in Lean in [8].
(O3)	`ramanujan_identity_five`	Ramanujan's identity (Remark 4.5). Modular forms on $\Gamma_0(5)$, or a q-series proof.
(O4)	`hardy_ramanujan_asymptotic`	Theorem 5.1. Circle method with (O5), (O6); or (O9), (O10) and a bound for the terms $k \geq 2$.
(O5)	`partitionFunction_eq_circleIntegral`	Cauchy's formula for $p(n)$. (O1) and Cauchy's integral formula.
(O6)	`partitionGenFun_asymptotic_near_one`	$P(e^{-t}) \sim \sqrt{t/2\pi}\, e^{\pi^2/6t}$ as $t \to 0^+$. (O1) and $\eta(-1/\tau) = \sqrt{-i\tau}\,\eta(\tau)$ from Mathlib.
(O7)	`eta_modular_transformation`	Transformation law (6.3) of η. Dedekind's functional equation [2, Ch. 3].
(O8)	`abs_rademacherTerm_le`	$\lvert T_k(n)\rvert \leq C_n\, k^{-3/2}$. Elementary, from Proposition 6.1 and (6.1).
(O9)	`tsum_rademacherTerm`	$\sum_{k\geq 1} T_k(n) = p(n)$. (O5), (O7), Farey dissection, Bessel integral [2, Ch. 5].
(O10)	`rademacherTerm_one_isEquivalent`	$T_1(n) \sim$ right-hand side of (1.3). Elementary real analysis.

Table 2: The ten statements whose proofs are `sorry`.

Audit. The dependencies in Table 1 are reported by Lean itself. For example (answers of Lean shown as comments):

```
#print axioms PartitionTheory.partition_generating_function
-- 'PartitionTheory.partition_generating_function' depends on axioms:
--   [propext, Classical.choice, Quot.sound]
#print sorries PartitionTheory.ramanujan_congruence_5
-- PartitionTheory.qPochhammerInfinite_pow_three has sorry of type
--   PartitionTheory.qPochhammerInfinite R ^ 3 = PartitionTheory.jacobiCubeSeries R
```

The three axioms in the first answer are the standard axioms of Lean and Mathlib. We ran `#print axioms` on all 50 theorems. Exactly 33 of them depend on these three axioms only; among them are Theorem 3.3, Lemmas 4.2, 4.4 and 5.2, Propositions 6.1 and 6.2, and the derivative formula (6.1). The other 17 depend in addition on `sorryAx`: they are the ten statements of Table 2 and the seven theorems proved relative to them, namely `ramanujan_congruence_5` with its two auxiliary lemmas on coefficients, `hasSum_partitionFunction_qParam`, `hardy_ramanujan_tendsto`, `summable_rademacherTerm` and `rademacher_exact_series`.

Numerical evidence. A false statement closed by `sorry` would invalidate everything derived from it. As a safeguard, the statements (O2)–(O4), (O6), (O7), (O9) and (O10) were tested

numerically with a transcription of the Lean definitions into Python; (O1) and (O5) are standard facts, and (O8) is proved informally in Section 6. For $1 \le n \le 100$ the sum $\sum_{k \le 200} T_k(n)$ differs from $p(n)$ by less than $2 \cdot 10^{-4}$, in accordance with (O9). The identities (O2) and (O3) hold up to X^{200} and X^{100}, respectively, in exact integer arithmetic. The transformation law (O7) holds with relative error below 10^{-8} for eight matrices and three points τ each. The quotient in (O4) is 0.9563, 0.9860 and 0.9956 for $n = 10^2, 10^3, 10^4$, and the quotients in (O6) and (O10) likewise approach 1. This is evidence that the statements are the intended ones; it is not a proof.

8 Conclusion

We have formalized in Lean 4 the definitions and statements of four classical theorems on the partition function. Euler's generating function is proved, Ramanujan's congruence modulo 5 is reduced to Jacobi's identity, and for the Hardy–Ramanujan formula and Rademacher's series the parts of the proofs that are independent of the circle method are verified. The remaining work falls into four groups of increasing difficulty.

1. (O1), (O5), (O6), (O8) and (O10) are real and complex analysis within reach of current Mathlib, which already contains Cauchy's integral formula and the transformation of η under $\tau \mapsto -1/\tau$.
2. (O2) should follow from the Jacobi triple product formalized in [8]. This would complete the proof of Theorem 4.1; the same method, with $E^6 = (E^3)^2$, gives $p(7n + 5) \equiv 0 \pmod 7$ [7].
3. (O7), (O9) and (O4) constitute the circle method proper. They require the transformation law of η with its Dedekind-sum multiplier, a Farey dissection of the circle or Ford circles, and an integral representation of $I_{3/2}$, none of which is in Mathlib at present. The Isabelle/HOL development [5] is a guide.
4. (O3) requires either the theory of modular forms on $\Gamma_0(5)$ or a longer q-series argument.

Acknowledgements

This work would not have been possible without the Mathlib library and its contributors. The Lean development and this text were modified with the assistance of the AI system Claude (Anthropic). Every statement described as proved has been checked by the Lean kernel, and the author takes responsibility for the content of the paper.

References

[1] G. E. Andrews, *The Theory of Partitions*, Encyclopedia of Mathematics and its Applications, vol. 2, Addison–Wesley, Reading, MA, 1976; reissued by Cambridge University Press, 1998.

[2] T. M. Apostol, *Modular Functions and Dirichlet Series in Number Theory*, 2nd ed., Graduate Texts in Mathematics, vol. 41, Springer, New York, 1990.

[3] M. Eberl, The partition function and the pentagonal number theorem, *Archive of Formal Proofs*, April 2025, https://isa-afp.org/entries/Pentagonal_Number_Theorem.html.

[4] M. Eberl, A. Bordg, L. C. Paulson and W. Li, Dedekind sums, *Archive of Formal Proofs*, April 2025, https://isa-afp.org/entries/Dedekind_Sums.html.

[5] M. Eberl and L. C. Paulson, Rademacher's series for the partition function, *Archive of Formal Proofs*, July 2026, https://isa-afp.org/entries/Rademacher_Series.html.

[6] G. H. Hardy and S. Ramanujan, Asymptotic formulae in combinatory analysis, *Proc. London Math. Soc.* (2) **17** (1918), 75–115.

[7] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., revised by D. R. Heath-Brown and J. H. Silverman, Oxford University Press, Oxford, 2008.

[8] K. Lau, S. Lee and K. Ono, Formalized q-series: the Rogers–Ramanujan identities and beyond, preprint, arXiv:2607.01544, 2026.

[9] D. H. Lehmer, On the Hardy–Ramanujan series for the partition function, *J. London Math. Soc.* **12** (1937), 171–176.

[10] D. H. Lehmer, On the series for the partition function, *Trans. Amer. Math. Soc.* **43** (1938), 271–295.

[11] D. H. Lehmer, On the remainders and convergence of the series for the partition function, *Trans. Amer. Math. Soc.* **46** (1939), 362–373.

[12] J. Liu, J. Zhang and L. Zhi, A formal proof of the irrationality of $\zeta(3)$ in Lean 4, preprint, arXiv:2503.07625, 2025.

[13] The mathlib Community, The Lean mathematical library, in: *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, ACM, 2020, pp. 367–381.

[14] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in: *Automated Deduction – CADE 28*, Lecture Notes in Computer Science, vol. 12699, Springer, 2021, pp. 625–635.

[15] H. Rademacher, On the partition function $p(n)$, *Proc. London Math. Soc.* (2) **43** (1937), 241–254.

[16] H. Rademacher, On the expansion of the partition function in a series, *Ann. of Math.* (2) **44** (1943), 416–422.

[17] S. Ramanujan, Some properties of $p(n)$, the number of partitions of n, *Proc. Cambridge Philos. Soc.* **19** (1919), 207–210.